All-n Formulae and Certified Advances for Difficult OEIS Sequences

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Abstract. We give exact all-dimension formulae or recurrences for five short and difficult entries of the On-Line Encyclopedia of Integer Sequences: A000530, A003167, A005787, A006545, and A007234. The methods combine run-composition counting, finite Egyptian-fraction recursion, affine-slice inclusion-exclusion, unlabelled cycle constructions, and weighted dynamic programming on permutation power-map functional graphs. The last method extends from squaring to every fixed map $\sigma \mapsto \sigma^d$. We also record exact terms and corrections for a further collection of OEIS entries. Each computational statement is paired with an explicit witness, an exhaustive partition of the search, or an exact arithmetic checker. In particular, the results include corrected values for A007234 and A002887, the exact value A003167(7) = 155068098, five exact values of A075099, and A337433(7) = 87.

Keywords: integer sequences; exact enumeration; Burnside's lemma; functional graphs; unlabelled generating functions; Egyptian fractions; computational certificates.

https://github.com/Siddhartha-Mahajan/oeis-sequences-audit

Contents

1 Introduction

Short integer sequences often sit at an awkward boundary between structural combinatorics and finite computation. A few initial terms may be accessible by direct enumeration, while the underlying objects admit symmetries, recursive constructions, or functional-graph decompositions that are not visible in the published data. The On-Line Encyclopedia of Integer Sequences (OEIS) [7] is particularly valuable at this boundary: it records the numerical problem, its historical sources, and the point at which a new theorem or a certificate can materially change the entry.

This paper develops five structural results and collects a larger set of exact computational advances. Each principal result determines its sequence for every positive integer n. The proofs use several different organizing ideas: safe-prefix trees, Egyptian-fraction recursion, affine patching over Boolean cubes, unlabelled rooted-tree species, and weighted independent sets in functional graphs.

1.1 Principal results

`Entry`	Structural result	Consequences and checks
`A000530`	Finite binomial sum obtained from ordered run compositions in a safe-prefix tree.	Exact state dynamic program through $n = 50$ and literal word enumeration for small n.
`A003167`	Finite recursion for nondecreasing solutions of $\sum_i 1/x_i = 1/2$, with an exact divisor terminal.	$a(7) = 155068098$, certified by 262 disjoint branches.
`A005787`	Inclusion-exclusion recurrence from the last OR operation, with an EGF functional equation and product-sum solution.	$a(6) = 323041664$, $a(7) = 284820663040$, $a(8) = 578706325429760$.
`A006545`	OGF for stable unlabelled unicyclic graphs from rooted-tree series and a complete transposition classification.	Exact coefficients to arbitrary order; independent graph enumeration through 12 vertices.
`A007234`	Weighted conjugacy-type recurrence for maximum squaring-free subsets of S_n.	Corrected $a(7) = 3390$, $a(8) = 29409$; extension to every fixed power map $\sigma \mapsto \sigma^d$.

Table 1: The all-dimension results proved in this paper.

The all-n statements are mathematical identities or finite recurrences; their validity does not depend on a numerical cutoff. Computation is used to evaluate the formulae, to verify implementations by structurally independent methods, and to certify selected finite terms.

1.2 Exact results and certificate types

The accompanying exact terms include minimally 2-edge-connected graphs, cutting centers of trees, free-monoid multiplication programs, triangular lattice labelings, simultaneous representations by two squares and two cubes, trapped knight paths, polygonal-number subset sums, and Hankel determinant extrema.

Four certificate patterns recur throughout the paper.

1. *Constructive witnesses.* A compact object is checked directly, such as a word, graph, labeling, or arithmetic representation.
2. *Disjoint exhaustive partitions.* A large search is split into branches whose union and pairwise disjointness are verified separately.
3. *Canonical exhaustive enumeration.* Every candidate is generated, a local defining property is

checked, and canonical labeling removes isomorphic duplicates.

4. *Optimization-assisted constructions.* The full integer model is included, while an independent checker verifies each displayed object. Exact optimality is asserted only when the included model has a completed global optimization record.

These patterns separate the mathematical reduction from the expensive search. They also make clear whether a statement is an identity, an exact finite term, a correction, or a convention-sensitive discrepancy.

1.3 Notation and repository organization

Partitions are written either as tuples or in frequency notation $\lambda = 1^{m_1}2^{m_2}\cdots$, with

$$z_\lambda = \prod_{j\geq 1} j^{m_j} m_j!.$$

For a formal power series F, the notation $[x^n]F$ denotes coefficient extraction. All generating-function identities are identities of formal power series, so no analytic convergence is required. Graphs are finite and simple unless otherwise stated.

Every sequence identifier in the paper links to its OEIS entry. Code, certificates, ancillary tables, and concise proposed OEIS edits are organized by identifier in the repository `https://github.com/Siddhartha-Mahajan/oeis-sequences-audit`. The principal result classes use the following paths:

all-dimension theorems: `entries/all_n/AXXXXXX/`;

exact finite determinations: `entries/exact_terms/AXXXXXX/`;

convention audits: `entries/bound_progress/AXXXXXX/`.

2 Run-composition formula for `A000530`

For a binary word w, let $c_s(w)$ be the number of occurrences of $s \in \{0, 1\}$, and let $m_s(w)$ be the length of its longest run of s. The sequence `A000530` counts first-hit words ending in zero: no proper prefix satisfies

$$c_0 + m_0 \geq 2n \qquad \text{or} \qquad c_1 + m_1 \geq 2n,$$

and the full word satisfies the zero condition. Put $N = 2n - 1$. A word is *safe* when

$$c_s(w) + m_s(w) \leq N \qquad (s = 0, 1). \tag{1}$$

For $r \geq 1$, define

$$q_N(r) = \#\left\{(x_1, \dots, x_r) \in \mathbb{Z}_{>0}^r : \sum_{i=1}^{r} x_i + \max_i x_i \leq N\right\}, \tag{2}$$

and set $q_N(0) = 1$.

Theorem 2.1 (Run-composition formula)**.** *For every $n \geq 1$,*

$$\boxed{a(n) = 1 + \sum_{r\geq 1} q_N(r)^2 + \sum_{r\geq 0} q_N(r)q_N(r+1), \qquad N = 2n - 1.} \tag{3}$$

Both sums are finite; in fact $q_N(r) = 0$ for $r \geq N$.

Proof. Safe words form the internal vertices of a finite rooted binary tree, with the empty word at the root. Appending either bit to a safe word either remains safe or produces a first hit. If the tree has S_n internal vertices, then it has $S_n + 1$ terminal children. Bit complementation is a fixed-point-free bijection between terminal words whose first hit is caused by zero and those whose first hit is caused by one. Therefore

$$a(n) = \frac{S_n + 1}{2}. \tag{4}$$

A symbol appearing in exactly r runs has an ordered run-length vector $(x_1, \ldots, x_r)$. Condition equation (1) for that symbol is precisely the inequality in equation (2). A nonempty binary word has either the same number r of zero-runs and one-runs, with two choices of first symbol, or $r + 1$ runs of one symbol and r of the other, with two choices for the symbol having more runs. Hence

$$S_n = 1 + 2\sum_{r\geq 1} q_N(r)^2 + 2\sum_{r\geq 0} q_N(r)q_N(r+1).$$

Substitution in equation (4) proves equation (3).

Finally, every positive r-tuple has sum at least r and maximum at least one, so $q_N(r) = 0$ when $r + 1 > N$; the slightly weaker statement $q_N(r) = 0$ for $r \geq N$ suffices. □

2.1 Explicit finite binomial sum

For integers r, m, T, define

$$B(r, m, T) = \sum_{j=0}^{r} (-1)^j \binom{r}{j}\binom{T - jm}{r}, \tag{5}$$

where $\binom{u}{r} = 0$ if $u < r$ or $u < 0$.

Proposition 2.2. *For $r \geq 1$,*

$$q_N(r) = \sum_{m=1}^{\lfloor (N-r+1)/2 \rfloor} \left(B(r, m, N - m) - B(r, m - 1, N - m)\right). \tag{6}$$

Consequently, equations (3), (5) and (6) form an explicit finite binomial-sum formula for $a(n)$.

Proof. Fix a possible maximum part m. The number of positive r-tuples with all parts at most m and total sum at most T equals $B(r, m, T)$. Indeed, subtract one from each positive part, introduce a nonnegative slack variable for the unused total, and apply inclusion-exclusion to the r upper bounds. The difference $B(r, m, N - m) - B(r, m - 1, N - m)$ therefore counts the tuples with maximum exactly m and total sum at most $N - m$. A positive r-tuple with maximum exactly m has total sum at least $m + r - 1$. Hence it can occur only when $2m + r - 1 \leq N$, or $m \leq \lfloor (N - r + 1)/2 \rfloor$. Summing over m gives equation (6). □

2.2 Finite-state cross-check

The state

$$(c_0, c_1, m_0, m_1, \text{last symbol, current run length})$$

determines every future transition and whether a predicate first becomes true. Equal states may therefore be merged with integer multiplicity. The dynamic program is finite: if both symbols occur in a safe word, then $c_s \leq 2n - 2$ for each s, so its length is at most $4n - 4$; constant safe words are shorter. Thus exhausting the safe frontier is exact rather than cutoff-based.

The run formula and the independent state dynamic program agree through $n = 50$, while literal word enumeration verifies the first five terms. The formula begins

$$\begin{gathered}1, 5, 28, 226, 2077, 20770, 222884, 2529541,\\ 29995812, 368025335, 4637331008, 59687103424,\\ 781671686963, 10386703092609, 139745901077363, \ldots.\end{gathered}$$

The stored formula sample continues through $n = 100$; the all-n formula itself has no computational endpoint.

3 An Egyptian-fraction recursion for A003167

An integral n-dimensional cuboid with nondecreasing side lengths $x_1 \le \cdots \le x_n$ has volume $\prod_i x_i$. Its total facet hyperarea is $2\sum_i \prod_{j\ne i} x_j$. Equality of volume and total facet hyperarea is therefore equivalent to

$$\frac{1}{x_1} + \cdots + \frac{1}{x_n} = \frac{1}{2}. \tag{7}$$

Thus A003167 counts nondecreasing positive-integer solutions of equation (7).

For coprime integers $p \ge 0$, $q \ge 1$, an integer $r \ge 0$, and a lower bound $L \ge 1$, let

$$C(r, p, q, L)$$

be the number of nondecreasing r-tuples $L \le x_1 \le \cdots \le x_r$ whose reciprocal sum is p/q. Use the terminal conventions

$$C(0, 0, 1, L) = 1, \qquad C(0, p, q, L) = 0 \quad (p > 0),$$

and $C(r, 0, 1, L) = 0$ for $r > 0$. For $r = 1$, the value is one exactly when $p > 0$, $p \mid q$, and $q/p \ge L$.

Theorem 3.1 (Exact finite recursion)**.** *For $r \ge 2$ and $p > 0$,*

$$C(r, p, q, L) = \sum_{x=\max\{L, \lfloor q/p\rfloor+1\}}^{\lfloor rq/p\rfloor} C(r-1, p_x, q_x, x), \tag{8}$$

where p_x/q_x is the reduced form of

$$\frac{px - q}{qx}. \tag{9}$$

In particular,

$$\boxed{\mathit{A003167}(n) = C(n, 1, 2, 1)}. \tag{10}$$

Proof. Suppose $x = x_1$ is the first denominator. Since $r - 1$ positive reciprocals remain, the residual sum must be positive, so $1/x < p/q$ and therefore $x > q/p$. Monotonicity also requires $x \ge L$. On the other hand every reciprocal is at most $1/x$, whence $p/q \le r/x$, or $x \le rq/p$. These are exactly the finite bounds in equation (8). After choosing x, subtracting $1/x$ produces equation (9), and the remaining denominators are at least x. Every solution has a unique first denominator and every summand constructs valid solutions, proving the recursion. Equation equation (10) is the reformulation equation (7). □

3.1 Two-denominator divisor terminal

When two denominators remain, the equation

$$\frac{1}{x} + \frac{1}{y} = \frac{p}{q}$$

is equivalent to

$$(px - q)(py - q) = q^2. \tag{11}$$

Hence the terminal count can be obtained by enumerating divisors $d \mid q^2$ and testing

$$x = \frac{d + q}{p}, \qquad y = \frac{q^2/d + q}{p}, \qquad L \le x \le y. \tag{12}$$

This is an exact acceleration of equation (8), not an added assumption. The implementation uses deterministic integer factorization and protects the products in q^2 with widened integer arithmetic.

3.2 The seventh term

Exact evaluation

$$\boxed{\texttt{A003167}(7) = 155068098.}$$

For $n = 7$, the first denominator lies in $3 \le x_1 \le 14$. The expensive branch $x_1 = 3$ was refined into the following disjoint pieces, each dictated by the same denominator bounds:

$$\begin{aligned}
&x_1 = 4, \ldots, 14; \\
&x_1 = 3, \quad x_2 = 8, \ldots, 36; \\
&x_1 = 3, \quad x_2 = 7, \quad x_3 = 44, \ldots, 210; \\
&x_1 = 3, \quad x_2 = 7, \quad x_3 = 43, \quad x_4 = 1807, \ldots, 7224,
\end{aligned}$$

with the last interval partitioned into consecutive subranges. The aggregation checker verifies the embedded prefixes and ranges, proves that the x_4-subranges cover $[1807, 7224]$ without gaps or overlaps, and sums 262 branch counts. Altogether the branches visit 151379115 recursion nodes and make 151271985 terminal pair calls.

The same C++ recursion reproduces the known values

$$2,\ 10,\ 108,\ 2892,\ 270332$$

for $n = 2, \ldots, 6$. A separate Python implementation using exact rational recursion, without the factorization terminal, independently reproduces this prefix. The theorem is all-dimensional; the branch computation supplies the new seventh numerical value.

4 Affine-slice recurrence for A005787

Consider an arrangement of n circles and identify its 2^n regions with $\{0, 1\}^n$, where x_i records membership in circle i. A black-white coloring is a Boolean function. Starting from the all-white coloring, the allowed operation on coordinate i either colors the whole circle black or reverses every color inside it. Algebraically these are

$$f \longmapsto f \vee x_i, \qquad f \longmapsto f \oplus x_i, \tag{13}$$

where arithmetic for $\oplus$ is over $\mathbb{F}_2$. This is the subset-coloring problem of Rubinstein, Shallit, and Szegedy [8].

Let F_n be the family of reachable functions and put $a(n) = |F_n|$, with the auxiliary value $a(0) = 1$. Let Aff_n be the 2^{n+1} affine functions on n variables.

4.1 The last-OR decomposition

For $0 \le i < n$, define

$$E_i = \left\{ f : f|_{x_i=0} \in F_{n-1} \text{ and } f|_{x_i=1} \in \mathrm{Aff}_{n-1} \right\}. \tag{14}$$

Lemma 4.1. *One has*

$$F_n = \bigcup_{i=0}^{n-1} E_i, \qquad |E_i| = 2^n a(n-1). \tag{15}$$

Proof. Choose the last OR operation in a construction. If it is performed on coordinate i, subsequent operations are XORs by coordinate functions. The $x_i = 0$ slice is therefore reachable in $n-1$ variables and the $x_i = 1$ slice is affine. A construction using no OR is itself affine and belongs to every E_i. Hence $F_n \subseteq \bigcup_i E_i$.

Conversely, suppose the zero-slice is $g \in F_{n-1}$ and the one-slice is $h = c \oplus \ell$, where ℓ is linear in the remaining variables. Reachability is closed under XOR by a linear form. Construct $g \oplus \ell$, apply OR on coordinate i, and then XOR by ℓ. The two slices are now g and $1 \oplus \ell$. If $c = 0$, one final XOR by x_i changes only the one-slice and gives ℓ; if $c = 1$, no final step is needed. Thus every element of E_i is reachable.

There are $a(n-1)$ choices for the zero-slice and 2^n affine functions on the remaining $n-1$ variables, proving the cardinality statement. □

Restriction to any collection of zero-coordinate faces preserves reachability: each operation restricts either to the corresponding operation in fewer variables or to the identity.

4.2 Multiple intersections

Lemma 4.2 (Affine patching)**.** *For every $I \subseteq \{0, \ldots, n-1\}$ with $|I| = k \ge 2$,*

$$\left| \bigcap_{i \in I} E_i \right| = 2^{n+1} a(n-k). \tag{16}$$

Proof. Take $f \in \bigcap_{i \in I} E_i$. For each $i \in I$, the slice $h_i = f|_{x_i=1}$ is affine, and the slices agree on every pairwise overlap. Any two compatible affine slices glue to a global affine function: their constant and linear coefficients agree on the common codimension-two face, and the two remaining coordinate coefficients are fixed by the respective slices. Let h be the affine function obtained from two slices. For a third $k \in I$, the difference $h|_{x_k=1} - h_k$ is affine on that slice and vanishes on its intersections with two distinct coordinate hyperplanes. The zero set of a nonzero affine function over $\mathbb{F}_2$ is a single affine hyperplane, so this difference is zero. Repeating the argument shows that a unique global affine function h agrees with f wherever at least one selected coordinate is one.

On the remaining face

$$Z_I = \{x : x_i = 0 \text{ for every } i \in I\},$$

put $q = f|_{Z_I}$. By restriction, $q \in F_{n-k}$. Hence f determines a pair

$$(h, q) \in \mathrm{Aff}_n \times F_{n-k}.$$

Conversely, given such a pair, define $f = h$ off Z_I and $f = q$ on Z_I. We prove by induction on k that $f \in \bigcap_{i \in I} E_i$. Fix $i \in I$. The one-slice is affine. Its zero-slice is the analogous patching construction in one fewer variable with selected set $I \setminus \{i\}$. When this set has one element, the converse part of lemma 4.1 shows that the slice is reachable. For larger sets, the induction hypothesis shows that the slice lies in the corresponding intersection and hence is reachable. Thus $f \in E_i$ for every $i \in I$. The two constructions are inverse bijections. Since $|\mathrm{Aff}_n| = 2^{n+1}$, equation (16) follows. □

4.3 Recurrence and generating function

Theorem 4.3 (All-n recurrence)**.** *With $a(0) = 1$,*

$$\boxed{a(n) = n2^n a(n-1) + 2^{n+1} \sum_{k=2}^{n} (-1)^{k+1} \binom{n}{k} a(n-k).} \tag{17}$$

Proof. Apply inclusion-exclusion to $F_n = \bigcup_i E_i$. The singleton terms use $|E_i| = 2^n a(n-1)$, and every k-fold intersection with $k \geq 2$ has the cardinality in equation (16). There are $\binom{n}{k}$ intersections of size k, giving equation (17). □

Let

$$A(x) = \sum_{n \geq 0} a(n) \frac{x^n}{n!}.$$

Multiplying equation (17) by $x^n/n!$, summing over n, and collecting the binomial convolution gives the formal functional equation

$$\boxed{A(x) = 1 + 2\big(1 - x - e^{-2x}\big)A(2x).} \tag{18}$$

Set $C(x) = 2(1 - x - e^{-2x})$. Since $C(0) = 0$, iterating equation (18) is coefficientwise legitimate in $\mathbb{Q}[[x]]$ and yields

$$\boxed{A(x) = \sum_{m \geq 0} \prod_{j=0}^{m-1} C(2^j x) = \sum_{m \geq 0} 2^m \prod_{j=0}^{m-1} \left(1 - 2^j x - e^{-2^{j+1} x}\right).} \tag{19}$$

The empty product is one, and the m-th summand has valuation at least m, so only finitely many summands affect any coefficient.

The recurrence begins

$$2,\ 8,\ 112,\ 5856,\ 869824,\ 323041664,\ 284820663040,\ 578706325429760, \ldots.$$

An explicit construction of all small intersection sets verifies equation (16) through four variables. A separate C++ enumerator independently obtains the complete six-circle count and its intersection table. Exact recurrence arithmetic through $n = 20$ agrees with the preserved certificate.

5 Stable unlabelled unicyclic graphs: `A006545`

A graph is semistable at a vertex v when every automorphism of $G - v$ preserves the former neighborhood of v, and is stable when its vertices admit a deletion ordering in which the current graph is semistable at the next vertex. McAvaney, Grant, and Holton proved that a connected unicyclic graph is stable if and only if its automorphism group contains a transposition [4, 5]. This characterization turns the enumeration into a local classification problem.

5.1 Rooted-tree series and decorated cycles

Let $R(x)$ be the ordinary generating function for unlabelled rooted trees. The classical Pólya-Otter equation is

$$R(x) = x \exp\left(\sum_{j\geq 1} \frac{R(x^j)}{j}\right). \tag{20}$$

Let $B(x)$ count rooted trees in which no vertex has two or more leaf children. In the rooted-tree multiset construction, arbitrary multiplicity of leaf children contributes $(1-x)^{-1}$, while multiplicity zero or one contributes $1+x$. Replacing the former by the latter multiplies the rooted operator by $(1+x)(1-x) = 1 - x^2$, giving

$$B(x) = x(1 - x^2) \exp\left(\sum_{j\geq 1} \frac{B(x^j)}{j}\right). \tag{21}$$

For a series F with $F(0) = 0$, let $D(F)$ be the OGF for unoriented cycles of length at least three whose vertices carry F-objects. Burnside's lemma for the dihedral group gives

$$D(F) = \sum_{k\geq 3} \frac{\mathrm{Rot}_k(F) + \mathrm{Ref}_k(F)}{2k}, \tag{22}$$

$$\mathrm{Rot}_k(F) = \sum_{d|k} \varphi(d) F(x^d)^{k/d}, \tag{23}$$

$$\mathrm{Ref}_{2m+1}(F) = (2m + 1)F(x)F(x^2)^m, \tag{24}$$

$$\mathrm{Ref}_{2m}(F) = m\left(F(x)^2 F(x^2)^{m-1} + F(x^2)^m\right). \tag{25}$$

Every connected unicyclic graph has a unique cycle, with a rooted tree attached at each cycle vertex. Consequently $D(R)$ counts all connected unlabelled unicyclic graphs.

5.2 Where a transposition can occur

Lemma 5.1 (Twin classification)**.** *If a connected unicyclic graph has an automorphism that transposes exactly two vertices u, v, then the pair has exactly one of the following forms.*

1. *u, v are two leaf children of the same vertex;*
2. *u, v are the two bare vertices of a triangle;*
3. *u, v are opposite bare vertices of a 4-cycle.*

Conversely, each of these configurations yields a transposition.

Proof. A transposition fixing every other vertex implies

$$N(u) \setminus \{v\} = N(v) \setminus \{u\};$$

thus u, v are twins. Suppose first that they are nonadjacent. If they have one common neighbor, any additional incident edge would also be common and would create a second cycle, so they are sibling leaves. If they have two common neighbors, the four vertices form the unique cycle, which is a 4-cycle, and the transposed vertices are bare. Three common neighbors create at least two cycles.

If u, v are adjacent, connectedness of a nontrivial unicyclic graph forces a common neighbor. There can be exactly one: no common neighbor would isolate a K_2 component, while two would create more than one cycle. The pair is therefore the two bare vertices of a triangle. In all three cases the indicated swap visibly preserves adjacency and fixes every other vertex. □

5.3 The ordinary generating function

The difference $D(R) - D(B)$ counts exactly the unicyclic graphs containing a pair of sibling leaves. Among the remaining B-decorated cycles, a triangle transposition has two bare vertices and an arbitrary B-object at the third, contributing $x^2 B(x)$. A 4-cycle transposition has two opposite bare vertices; the other opposite pair carries an unordered pair of B-objects, whose OGF is

$$\frac{B(x)^2 + B(x^2)}{2}.$$

The triangle and 4-cycle families are disjoint and contain no sibling-leaf transposition.

Theorem 5.2 (All-n OGF)**.** *The ordinary generating function for stable connected unlabelled unicyclic graphs is*

$$\boxed{A(x) = D(R) - D(B) + x^2 B(x) + \frac{x^2}{2}\left(B(x)^2 + B(x^2)\right),} \tag{26}$$

where R, B, D are determined by equations (20) to (25). Thus $a(n) = [x^n]A(x)$ for every $n \geq 3$.

Proof. By the cited structural theorem, stability is equivalent to the existence of a transposition. Lemma 5.1 partitions such graphs into those with sibling leaves and the two exceptional bare-cycle mechanisms. The four terms in equation (26) count these disjoint classes exactly. □

The coefficients begin, with offset three,

$$\begin{aligned} &1, 2, 3, 8, 22, 62, 176, 500, 1425, 4078, \\ &\ 11666, 33447, 95922, 275332, 790518, 2270963, \\ &\ 6525674, 18759532, \ldots. \end{aligned}$$

The exact series code independently reproduces the total unicyclic counts through 12 vertices. Complete graph6 enumerations of connected unlabelled n-vertex, n-edge graphs were checked in three ways: by the recursive stability definition, by direct automorphism-group sizes at every deletion state, and by the elementary transposition test of lemma 5.1. All three select the same records through $n = 12$.

6 Power-map-free subsets of symmetric groups: `A007234`

Let F_n be the maximum cardinality of a subset $X \subseteq S_n$ such that

$$\sigma \in X \quad \Longrightarrow \quad \sigma^2 \notin X. \tag{27}$$

Equivalently, F_n is the independence number of the functional graph of the squaring map, with a looped vertex forbidden. Bouchard and Yeh studied this general f-free-set problem and reported the first values [2]. The conjugacy-type recurrence below corrects the last two of those values and extends the sequence to every n.

6.1 Cycle types and square-root multiplicities

Write a cycle type as $\lambda = 1^{m_1}2^{m_2}\cdots n^{m_n}$ and put

$$z_\lambda = \prod_{j=1}^{n} j^{m_j} m_j!. \tag{28}$$

The conjugacy class has size $n!/z_\lambda$. Let $q(\mu)$ be the cycle type obtained by squaring a permutation of type μ: an odd j-cycle remains a j-cycle, while an even $2j$-cycle becomes two j-cycles.

Lemma 6.1 (Uniform root fibers)**.** *If $q(\mu) = \lambda$, then every fixed permutation of type λ has exactly*

$$r(\lambda, \mu) = \frac{z_\lambda}{z_\mu} \tag{29}$$

square roots of type μ.

Proof. The squaring map sends the conjugacy class of μ equivariantly into the class of λ. Its image is nonempty and invariant under conjugation; hence it is the whole target class. Equivariance also makes every fiber have the same size. Dividing the source-class size $n!/z_\mu$ by the target-class size $n!/z_\lambda$ gives equation (29). □

6.2 Reverse trees

For a type λ containing an even part, let E_λ and I_λ be the maximum sizes of an independent set in the reverse tree below one permutation of type λ, conditional on excluding or including its root. These values depend only on the type: conjugation carrying one root permutation to another is an isomorphism of their rooted reverse functional graphs. Distinct square roots have disjoint reverse subtrees, because every permutation has a unique square. Therefore

$$E_\lambda = \sum_{q(\mu)=\lambda} r(\lambda, \mu) \max\{E_\mu, I_\mu\}, \tag{30}$$

$$I_\lambda = 1 + \sum_{q(\mu)=\lambda} r(\lambda, \mu) E_\mu. \tag{31}$$

These equations are acyclic. Let

$$v(\lambda) = \max\{v_2(j) : m_j(\lambda) > 0\}.$$

If λ has an even part and $q(\mu) = \lambda$, then $v(\mu) = v(\lambda) + 1$: a target cycle having maximal 2-adic valuation can only arise from a cycle twice as long, and no source cycle can have larger valuation without producing a larger target valuation. Processing types in descending v therefore determines all nonperiodic states. The reverse trees are finite because cycle lengths cannot be doubled indefinitely inside S_n.

If all parts of λ are odd, the same formulae determine the tree attached to a periodic cycle vertex after omitting the unique same-type root. Indeed, $q(\lambda) = \lambda$ and $r(\lambda, \lambda) = 1$; that root is the predecessor on the directed cycle, not a child in the attached tree.

6.3 Periodic conjugacy classes

For an all-odd type λ, put

$$o_\lambda = \operatorname{lcm}\{j : m_j(\lambda) > 0\}, \qquad L_\lambda = \operatorname{ord}_{o_\lambda}(2), \tag{32}$$

with $L_\lambda = 1$ when $o_\lambda = 1$. Every permutation in the class has order o_λ, so its squaring orbit has length L_λ. The class splits into

$$N_\lambda = \frac{n!/z_\lambda}{L_\lambda} \tag{33}$$

directed cycles.

Let $g_\lambda = \max\{0, I_\lambda - E_\lambda\}$. Excluding every cycle root contributes $L_\lambda E_\lambda$. Including a nonadjacent set of cycle roots gains g_λ per selected root, so a cycle of length greater than one contributes

$$L_\lambda E_\lambda + \left\lfloor \frac{L_\lambda}{2} \right\rfloor g_\lambda. \tag{34}$$

When $L_\lambda = 1$, the root has a loop and is forbidden, so the contribution is E_λ.

Theorem 6.2 (Conjugacy-type recurrence)**.** *Equations equations (29) to (34) form an exact finite recurrence for F_n. Explicitly,*

$$F_n = \sum_{\lambda \text{ all odd}} N_\lambda \begin{cases} E_\lambda, & L_\lambda = 1, \\ L_\lambda E_\lambda + \lfloor L_\lambda/2 \rfloor g_\lambda, & L_\lambda > 1. \end{cases} \tag{35}$$

Proof. Every finite functional graph is a disjoint union of directed cycles with rooted in-trees attached. The reverse-tree states equations (30) and (31) are the usual include/exclude dynamic program, with root multiplicities supplied by lemma 6.1. The valuation ordering proves that these states are computed without circular dependence away from all-odd types. On an all-odd class the squaring map is a permutation with cycle length L_λ, and the same-type predecessor has been removed from the attached tree. The maximum-weight independent set on each directed cycle is equation (34), with the loop exception at length one. Summing all components yields equation (35). □

The recurrence begins, with offset one,

$$\begin{aligned} &0,\ 1,\ 4,\ 16,\ 72,\ 522,\ 3390,\ 29409,\ 267561, \\ &\quad 2820600,\ 30658050,\ 377859960,\ 4866471720, \\ &\quad 70224099120,\ 1052687890800,\ 17121988170000, \ldots. \end{aligned}$$

Thus the values formerly reported as 3642 and 30753 at $n = 7, 8$ are corrected to

$$\boxed{F_7 = 3390, \qquad F_8 = 29409.} \tag{36}$$

The recurrence agrees through $n = 50$ with its stored exact table. Complete functional graphs through S_{11}, two independent Python component algorithms, and an exhaustive audit of every square-root fiber through S_7 give structurally independent checks. Explicit maximum-set rank lists for $n = 7, 8, 9$ certify the lower-bound side directly.

6.4 Every fixed power map

The argument extends without a change of principle to $P_d : S_n \to S_n$, $P_d(\sigma) = \sigma^d$, for any fixed $d \geq 2$. A cycle of length j becomes $\gcd(j, d)$ cycles of length $j/\gcd(j, d)$. Let $q_d(\mu)$ be the induced type map. Equivariance gives

$$r_d(\lambda, \mu) = \frac{z_\lambda}{z_\mu} \qquad (q_d(\mu) = \lambda). \tag{37}$$

Let $\mathcal{P}$ be the set of prime divisors of d, and define

$$h_d(\lambda) = \sum_{p\in\mathcal{P}} \max\{v_p(j) : m_j(\lambda) > 0\}. \tag{38}$$

For a source cycle of length ℓ and its target length $j = \ell/\gcd(\ell, d)$,

$$v_p(j) = \max\{v_p(\ell) - v_p(d), 0\}. \tag{39}$$

If a target type has a part not coprime to d, then every reverse type has strictly larger h_d: for at least one prime $p \mid d$, a positive target valuation forces the corresponding source valuation to increase by $v_p(d)$, while the maxima for other primes cannot decrease. The reverse type graph is therefore acyclic away from types whose parts are all coprime to d.

For such a periodic type, put

$$o_\lambda = \operatorname{lcm}\{j : m_j(\lambda) > 0\}, \qquad L_\lambda = \operatorname{ord}_{o_\lambda}(d). \tag{40}$$

The class splits into power-map cycles of length L_λ; omitting the unique same-type predecessor and applying the same weighted-cycle formula computes the largest $X \subseteq S_n$ satisfying

$$\sigma \in X \Longrightarrow \sigma^d \notin X.$$

This gives an exact all-n recurrence for every fixed d. The generic implementation specializes to theorem 6.2 for $d = 2$ and agrees with direct functional graphs for $d = 2, 3, 4, 6$ through S_7.

7 Selected exact finite determinations

The all-n results above are accompanied by exact finite determinations whose proofs combine a structural reduction with a complete certificate. This section records the strongest examples in enough detail to expose the exhaustiveness argument.

7.1 Cutting centers of trees: `A002887`

For a tree T of order N, delete a vertex v and let the component orders be $s_1, \ldots, s_d$. The cutting number is

$$c_T(v) = \sum_{i<j} s_i s_j = \frac{(N-1)^2 - \sum_i s_i^2}{2}. \tag{41}$$

Harary and Ostrand proved that the vertices of maximum cutting number form a path [3]. Suppose the intended center path is $v_1, \ldots, v_k$. Removing its path edges leaves rooted components of orders $b_1, \ldots, b_k$. Let q_i be the sum of the squared orders of the off-path branches incident with v_i, and put

$$L_i = \sum_{j<i} b_j, \qquad R_i = \sum_{j>i} b_j.$$

The path vertices have a common cutting number exactly when

$$L_i^2 + R_i^2 + q_i = K \qquad (1 \le i \le k) \tag{42}$$

for one integer K. Their common value is $((N-1)^2 - K)/2$.

For fixed N and proposed center value C, call a rooted tree of order s safe when every vertex in it has cutting number strictly below C after the root is attached to an outside component of order $N - s$. If the root has safe child-subtrees of orders $t_1, \ldots, t_d$, with $\sum t_i = s - 1$ and $q = \sum t_i^2$, its cutting number is

$$(N - s)(s - 1) + \frac{(s-1)^2 - q}{2}. \tag{43}$$

Thus safety is characterized recursively by an exact integer-partition dynamic program. This condition is necessary by deleting the root and sufficient by attaching witnessing safe subtrees. Enumerating positive compositions $(b_1, \ldots, b_k)$, exact attainable square sums in equation (42), and the safe-subtree spectra therefore loses no trees.

Searching orders increasingly gives

k	1	2	3	4	5	6	7	8	9	10
minimum N	3	4	7	10	48	48	122	122	264	264

The value at $k = 5$ corrects 50 to 48. Every accepted profile is realized as an explicit edge list; an independent checker deletes every vertex, recomputes all component orders and cutting numbers, and verifies that the maximizers are exactly the intended path.

7.2 Three square and three cube representations: A273354

Let $r_2^+(N)$ and $r_3^+(N)$ be the numbers of unordered positive pairs representing N as a sum of two squares and two cubes. The exact third term is

$$\boxed{\text{A273354}(3) = 11177126654841000000.} \tag{44}$$

Put $N_0 = 11177126654841000000 = 3343221000^2$. Direct integer arithmetic gives exactly the three cube representations

$$\begin{aligned} N_0 &= 279300^3 + 2234400^3 \\ &= 790020^3 + 2202480^3 \\ &= 1256850^3 + 2094750^3. \end{aligned}$$

Its factorization is

$$N_0 = 2^6 3^6 5^6 7^6 19^4.$$

The sum-of-two-squares divisor formula gives exactly three positive unordered square representations; the certificate lists them explicitly.

For minimality, suppose $M < N_0$ has at least three positive-cube representations. Dividing all six coordinates by their common gcd writes $M = g^3 m$, where m is a primitive three-cube value. Thus every candidate occurs among the cubic multiples below N_0 of the primitive values in the complete A003825 ancillary list. There are 110969 distinct candidates. Each was factored exactly using deterministic Miller-Rabin and Pollard-rho arithmetic, and the positive-square-pair formula was evaluated from the prime factorization. None has $r_2^+(M) = 3$. An independent verifier regenerates the candidate set, checks primality and factor products, recomputes the square counts, and directly scans all cube pairs of N_0.

7.3 Missing polygonal cases: A363253

For $s \geq 3$, let

$$P_s(k) = \frac{(s-2)k^2 - (s-4)k}{2}$$

be the k-th positive s-gonal number. Representations are sums of distinct positive s-gonal numbers, including the singleton representation. The missing cases are

$$\boxed{\texttt{A363253}(6) = \texttt{A363253}(7) = -1.} \tag{45}$$

The proof uses a conductor lemma. If subset sums of $P_s(1), \dots, P_s(m)$ cover every integer in $[L, U]$ and $P_s(m+1) \le U - L + 1$, then adjoining $P_s(m+1)$ makes the old and shifted intervals overlap. Since the successive polygonal numbers in the relevant range grow by a factor less than two, the overlap persists indefinitely. Exact bitset calculations show that

$$P_6(1), \dots, P_6(11) \text{ cover } [268, 678],$$
$$P_7(1), \dots, P_7(13) \text{ cover } [388, 1523].$$

Therefore every integer at least 268 is a distinct hexagonal subset sum and every integer at least 388 is a distinct heptagonal subset sum.

For $k \ge 68$, the six differences $P_6(k) - P_6(k-d)$, $1 \le d \le 6$, lie beyond the conductor and below $P_6(k-d)$. Appending the distinct largest summands $P_6(k-d)$ gives six non-singleton representations, hence at least seven in total. The analogous seven differences for heptagonal numbers work for $k \ge 79$, giving at least eight representations. Exact finite-prefix subset DP rules out the desired multiplicities before these thresholds.

7.4 Minimally 2-edge-connected graphs: A001072

The exact new term is

$$\boxed{\texttt{A001072}(15) = 41142.} \tag{46}$$

By the Whitney ear theorem [9], every 2-edge-connected graph is built from a cycle by adding ears. In a minimal 2-edge-connected simple graph, every later ear has a new internal vertex, and every prefix remains minimal. The generator recursively adds all open ears with at least one new internal vertex and all simple closed ears with at least two, retaining exactly the graphs that are connected and bridgeless but cease to be so after any edge deletion. Canonical graph6 records produced by nauty [6] remove isomorphic duplicates.

The computation gives 203421 accepted precanonical candidates and 41142 canonical records at order 15, after reproducing every published term through order 14. An independent verifier uses a different minimality test: for each edge e, it checks that $G - e$ is connected and finds an edge f for which $G - \{e, f\}$ is disconnected. A separate canonicalization pass leaves the count and record hash unchanged.

7.5 Free-monoid multiplication programs: A075099

Every binary word of length n requires one final multiplication. A shorter word contributes one additional multiplication exactly when it is retained as a reusable intermediate. Binary variables select these intermediates. For every selected word and every required target, split variables require a cut whose two factors are generators or selected shorter words. Since factors are strictly shorter, the local split conditions are equivalent to a valid multiplication program ordered by word length.

The resulting mixed-integer program minimizes 2^n plus the number of selected intermediates. Completed HiGHS runs have matching primal and global bounds and zero gap, giving

$$\boxed{a(7) = 151, \quad a(8) = 276, \quad a(9) = 556, \quad a(10) = 1066, \quad a(12) = 4171.} \tag{47}$$

The selected programs are checked independently by recursively verifying a valid factorization of every intermediate and every target. The stored time-limited run at $n = 11$ proves only $2118 \le a(11) \le 2138$; it is not used in equation (47).

7.6 Triangular-lattice labeling: A337433

For the 28 nodes of the order-seven triangular array, binary variables assign one label in $\{1, \ldots, 7\}$ to each node. If a node receives label k, the model requires an occurrence of every label $1, \ldots, k-1$ among its neighbors. This is exactly the defining local condition. A completed 53,115-node branch-and-bound computation has objective and global bound 87 with zero gap. The resulting labeling is independently checked node by node, and therefore

$$\boxed{\texttt{A337433}(7) = 87.} \tag{48}$$

7.7 Additional exact values

Entry	Exact result	Certificate
A323560	$a(19) = 36001752$	Exhaustive reversed trapped-knight paths, with a Held-Karp lower bound for the unvisited required neighbors.
A368353	$a(6) = -2579770$	Exhaustive $11!/2$ reversal classes of 6×6 Hankel matrices; exact fraction-free Bareiss determinants [1].
A368354	$a(6) = 2256911$	
A368355	$a(6) = 2579770$	

Table 2: Further exact finite determinations.

8 A convention-sensitive audit and reproducibility

8.1 The discrepancy in A323134

For A323134, an independent geometric enumerator agrees with the listed values through $n = 4$ and then gives

$$3034,\ 64877,\ 1503790 \qquad (n = 5, 6, 7),$$

instead of the first two listed values 3031 and 64866. The program counts simple closed straight-edge knight walks modulo translation, all eight square symmetries, cyclic starting point, and reversal, allowing collinear consecutive knight moves. A separate Python recount verifies 58840 rooted representatives and 3034 equivalence classes at $n = 5$. Because the historical convention on collinearity and equivalence may differ, this result is recorded as a definition-sensitive discrepancy rather than a proposed data correction.

8.2 Exact computational records

The exhaustive counting programs use integer or rational arithmetic. Large searches are divided into disjoint branches or preserved as canonical records whose coverage and aggregate counts are checked separately. The mixed-integer models for A075099 and A337433 contain only integral combinatorial data; their exactness claims use completed global optimization runs with zero gap, while separate solver-free checkers verify the displayed constructions. Incomplete optimization runs are not used as exactness claims.

C++ enumerators use deterministic seeds where search order matters. Each package states its mathematical reduction, replay command, and certificate boundary, and every distributed source, witness, and certificate is covered by the repository's SHA-256 manifest.

9 Conclusion

The principal results replace finite prefixes by structural descriptions. Safe binary words yield a finite run-composition formula for A000530; cuboids reduce to a finite Egyptian-fraction recursion for A003167; last-OR decomposition gives the recurrence and EGF of A005787; the transpositions of unicyclic graphs give the OGF of A006545; and conjugacy types compress the full power-map functional graph on S_n to a partition-indexed recurrence for A007234. The same functional-graph method applies to every fixed power d.

The finite results illustrate a complementary principle: an expensive computation becomes mathematically useful when its search space is completely specified and its conclusion is reduced to a compact, independently checkable certificate. Ear decompositions, cutting-center profiles, branch partitions, factorization records, and completed integer programs each provide such a reduction. Together, the formulae and certificates extend the underlying OEIS records while preserving a clear boundary between theorem, exact computation, and convention-sensitive enumeration.

A Complete result and certificate map

Every entry is stored under its OEIS identifier in `entries/all_n/` or `entries/exact_terms/`. The convention-sensitive A323134 audit remains under `entries/bound_progress/`. The table records the results treated in this article; the A368353–A368355 computation is stored in the A368355 folder.

Table 3: Result and certificate map for every entry in the archive.

Entry	Result class	Principal statement
A000530	all-n	Finite run-composition and binomial-sum formula; state-DP cross-check through $n = 50$.
A003167	all-n	Exact Egyptian-fraction recursion and $a(7) = 155068098$.
A005787	all-n	Exact recurrence, EGF equation, product-sum, and terms through the stored horizon.
A006545	all-n	Ordinary generating function for stable unlabelled unicyclic graphs.
A007234	all-n	Conjugacy-type recurrence; corrected $a(7) = 3390$, $a(8) = 29409$; every fixed power map.
A001072	exact	$a(15) = 41142$, minimally 2-edge-connected unlabelled graphs.
A002887	exact	Minimum tree orders for cutting-center paths of sizes 5 through 10: $48, 48, 122, 122, 264, 264$.
A273354	exact	$a(3) = 11177126654841000000$.
A323560	exact	$a(19) = 36001752$, trapped self-avoiding knight paths.
A363253	exact	$a(6) = a(7) = -1$, by conductor and finite-prefix arguments.
A368353--A368355	exact	At $n = 6$: $-2579770, 2256911, 2579770$, exact Hankel determinant extrema.
A075099	exact	$a(7) = 151$, $a(8) = 276$, $a(9) = 556$, $a(10) = 1066$, and $a(12) = 4171$.
A337433	exact	$a(7) = 87$, the minimum possible maximum edge difference in the triangular lattice graph.
A323134	convention audit	Independent counts $3034, 64877, 1503790$ at $n = 5, 6, 7$; no data edit asserted.

Each listed entry folder contains a mathematical note and a README. Where a direct OEIS submission is appropriate, concise proposed changes appear in an `OEIS_EDITS.md` or `OEIS_EDIT.md` file.

Witnesses and preserved outputs are under `certificates/`; regenerations and temporary binaries are kept under `build/`.

B Computational certificate details

B.1 Independent implementations for all-n formulae

For A000530, one implementation evaluates the binomial sum of theorem 2.1 and proposition 2.2; the other propagates compressed safe-prefix states. These methods share only the sequence definition and agree through $n = 50$. Literal uncompressed enumeration checks $n \leq 5$.

For A005787, the recurrence checker explicitly constructs every reachable function and every last-OR intersection through four variables. A separate compiled enumerator computes the complete six-circle reachable set and the intersection cardinalities used in inclusion-exclusion.

For A006545, the series implementation is checked against complete unlabelled graph files through 12 vertices. One verifier applies the recursive semistability definition, a second compares automorphism-group cardinalities, and a third tests the transposition classification of lemma 5.1 directly.

For A007234, the partition recurrence is compared with three complete functional-graph algorithms on permutations. The root-fiber checker exhausts all permutations through S_7, groups every square root by source and target cycle type, and verifies z_λ/z_μ for each nonempty fiber. The generic power-map verifier repeats the direct comparison for exponents $2, 3, 4, 6$.

B.2 The A003167 branch certificate

The large $n = 7$ Egyptian-fraction computation is archived as 262 text records. Every record stores its fixed denominator prefix, its permitted range for the next denominator, the branch count, the number of recursion nodes, and the number of terminal pair calls. The aggregation checker performs four independent tasks:

1. recompute the legal interval at each fixed prefix from equation (8);
2. verify that the top-level pieces cover $x_1 = 3, \ldots, 14$ exactly;
3. verify that the exceptional x_4-ranges at prefix $(3, 7, 43)$ are consecutive from 1807 through 7224;
4. sum the branch counts and diagnostics to the final total.

This separates the expensive enumeration from the inexpensive proof that no branch was omitted or counted twice. The final JSON certificate preserves the entire partition summary and its exact aggregate 155068098.

B.3 Cutting-center profile search

For A002887, the profile search does not enumerate unlabelled trees. It enumerates the complete numerical data forced by a center path: the positive composition (b_i), the off-path partition square sums q_i, and the common constant K in equation (42). The safe-rooted-subtree DP then proves realizability while excluding any off-path vertex tied with the center. The inequalities used to prune three consecutive path components are necessary conditions from the Harary–Ostrand analysis, not heuristic cuts. Accepted profiles are converted to concrete edge lists, so the final certificate checker is independent of the profile equations.

B.4 Canonical graph records

The A001072 archive contains all 41142 canonical graph6 records at order 15, a SHA-256 file, generator logs, and an independent verification log. The generator and verifier use different minimal 2-edge-connectivity tests. A second nauty canonicalization pass leaves both count and hash unchanged.

The A006545 graph files similarly separate all unicyclic records from the stable subset at each order. The certificate is therefore not only a count: it preserves the exact selected isomorphism classes and permits direct set comparison between alternative characterizations.

B.5 Arithmetic minimality for A273354

The minimality certificate records every cubic scaling of a primitive three-cube value below N_0, together with a complete prime factorization. The verifier does not trust the stored arithmetic. It checks the source primitive index, reconstructs each scaling, proves every recorded prime by deterministic 64-bit Miller–Rabin, multiplies the factorization back to the candidate, and recalculates r_2^+. It also directly scans the cube representations of N_0.

The only external completeness input is the primitive A003825 b-file through N_0. As an independent consistency test, cubic multiples of that list reproduce the first 100000 values of the supplied A018787 b-file without omissions or extras.

B.6 Exact optimization certificates

The A075099 and A337433 folders contain complete mixed-integer models and terminal solver records with coincident primal and dual bounds. The zero gaps certify global optimality. Independent solver-free checkers validate the intermediate-word programs for A075099 and the triangular-lattice labeling for A337433; these witnesses establish the matching feasible values without relying on the optimizer's internal object format.

For A075099, the exact certificates cover $n = 7, 8, 9, 10, 12$. The stored $n = 11$ run has a nonzero gap and is not used to assert an exact value. For A337433, the $n = 7$ record terminates with objective and best bound both equal to 87 after 53115 branch-and-bound nodes.

C Representative replay commands

Commands below are run from the repository root unless a subshell changes into an entry directory. Most quick verifiers use only the Python standard library. C++ sources use C++20 unless an entry README specifies C++17.

C.1 All-n results

```
(cd entries/all_n/A000530 && \
  python3 scripts/formula.py --max-n 50 \
    --compare certificates/dp_n1_n50.json && \
  python3 scripts/verify_bruteforce.py --max-n 5)
```

```
(cd entries/all_n/A003167 && \
  python3 scripts/aggregate_n7.py \
    --input-dir certificates/branches \
    --output build/n7_exact_certificate.json && \
  python3 scripts/verify_certificate.py \
    build/n7_exact_certificate.json)

(cd entries/all_n/A005787 && \
  python3 scripts/verify_recurrence.py && \
  python3 scripts/verify_small_and_arithmetic.py)

(cd entries/all_n/A006545 && \
  python3 scripts/generating_function.py \
    --max-n 100 --check \
    --bfile build/sample_n3_n100.txt)

(cd entries/all_n/A007234 && \
  python3 scripts/verify_recurrence.py --through 50 && \
  python3 scripts/verify_power_generalization.py)
```

C.2 Exact finite results

```
(cd entries/exact_terms/A001072 && \
  python3 scripts/verify_independent.py \
    certificates/n15_graphs.g6)

(cd entries/exact_terms/A002887 && \
  python3 scripts/replay_all.py --certificates-only)

(cd entries/exact_terms/A273354 && \
  python3 scripts/verify_minimality_certificate.py)

(cd entries/exact_terms/A363253 && \
  python3 scripts/prove_a6_a7.py \
    --output certificates/a6_a7_certificate.replay.json)

(cd entries/exact_terms/A075099 && \
  for f in certificates/n{7,8,9,10,12}_milp.json; do \
    python3 scripts/verify_construction.py "$f"; \
  done)

(cd entries/exact_terms/A337433 && \
  python3 scripts/verify_witness.py \
    certificates/n7_milp.json)

(cd entries/bound_progress/A323134 && \
  python3 scripts/verify_n5.py)
```

C.3 Archive integrity and manuscript build

```
sha256sum -c MANIFEST.sha256

cd manuscript
```

```
latexmk -pdf -interaction=nonstopmode main.tex
```

Data and code availability

All source code, exact certificates, witness files, ancillary values, proposed OEIS edits, and the LaTeX source of this manuscript are available in the public repository `https://github.com/Siddhartha-Mahajan/oeis-sequences-audit`. The archive includes a SHA-256 manifest covering every packaged source and certificate file.

Declaration on AI-assisted tools

AI-assisted software was used in exploratory computation, implementation, and editorial preparation. Every mathematical claim reported here is supported by the proofs or exact certificates described in the article and accompanying repository. The authors reviewed the resulting statements and take responsibility for the content of the work.

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